

Runtime adaptation as a programming pattern in service composition



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Coordination
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Take-away message

Take-away message

Runtime architectural adaptation

Take-away message

Runtime architectural adaptation
= Monitoring

Take-away message

Runtime architectural adaptation

= Monitoring

+ Decision policy

Take-away message

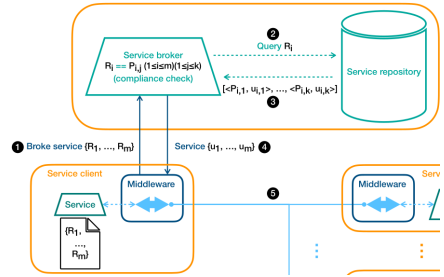
Runtime architectural adaptation

= Monitoring

+ Decision policy

+ Semantic-based service composition

from



Take-away message

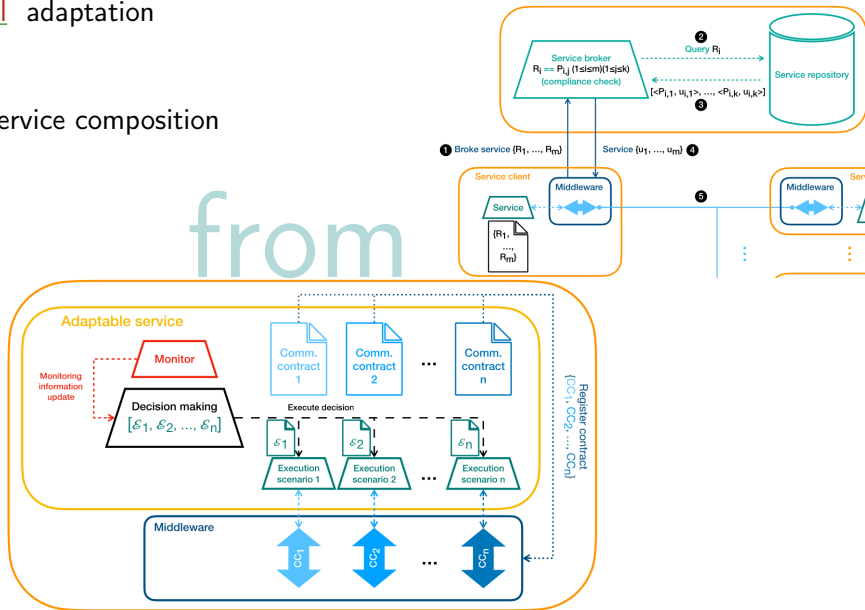
Runtime architectural adaptation

= Monitoring

+ Decision policy

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from
to



- ▶ A bird-eye watch of *SEArch*

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- ▶ Adaptation (as we see it)

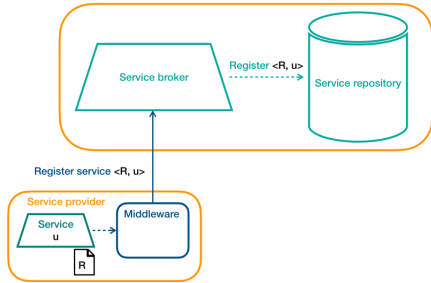
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- ▶ Adaptation (as we see it)
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- ▶ Some conclusions

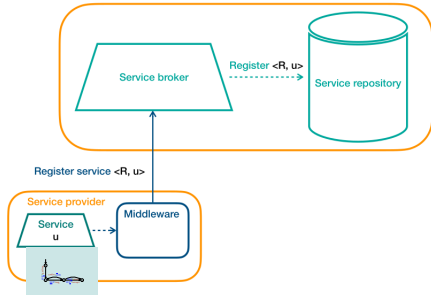
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– Prelude –

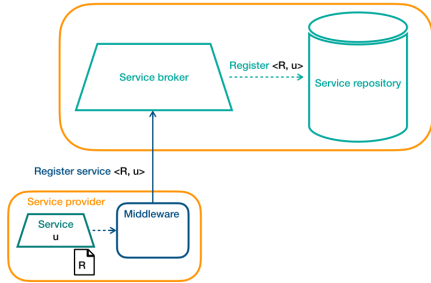
SEARCH basics



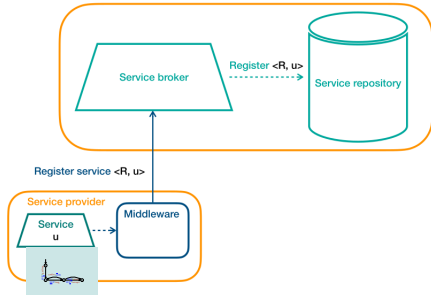
SEArch basics



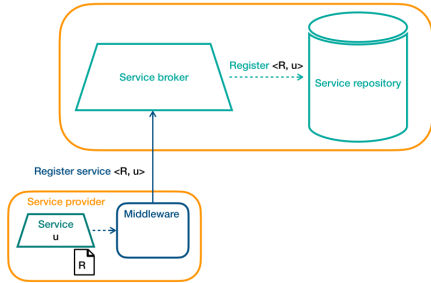
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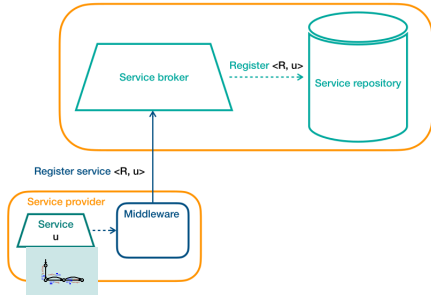
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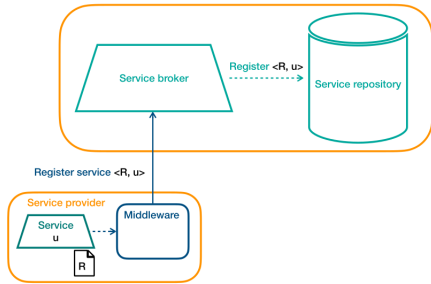
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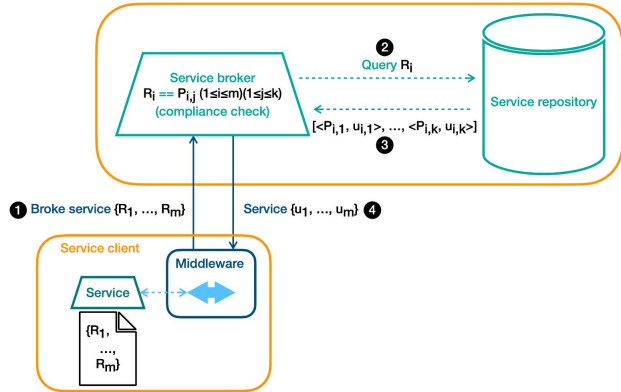
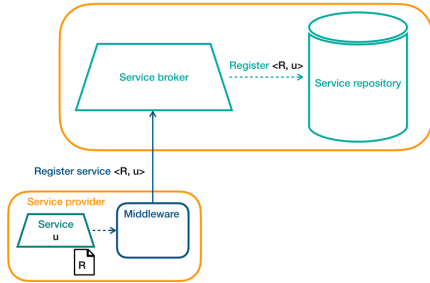
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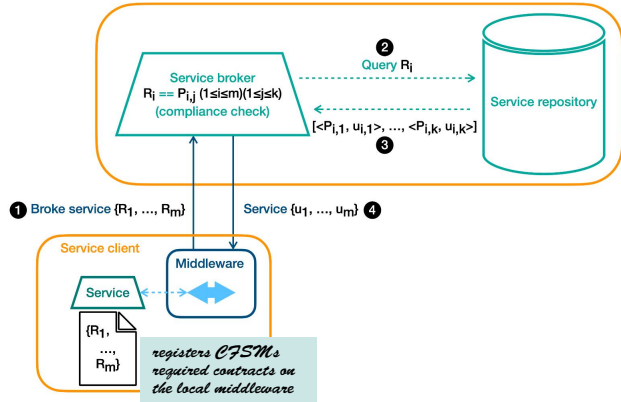
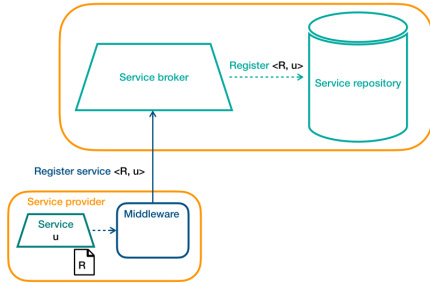
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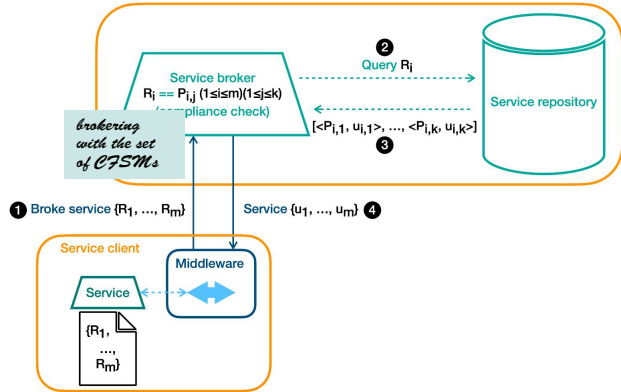
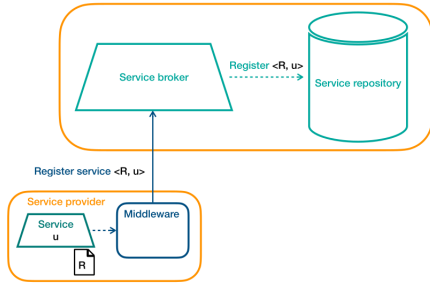
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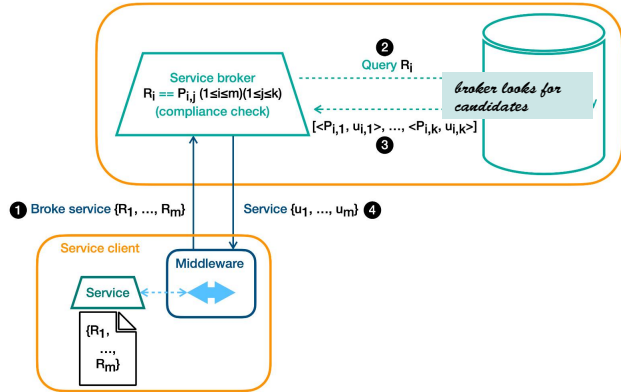
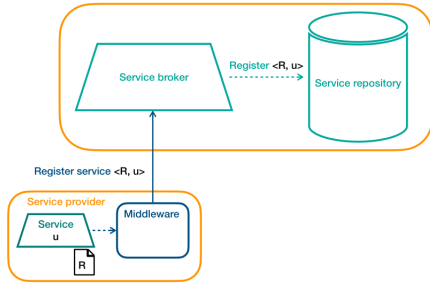
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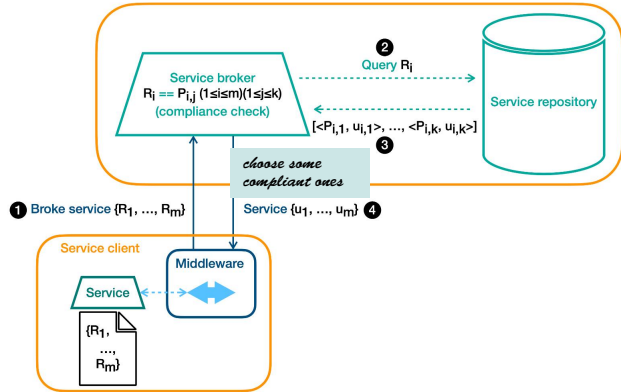
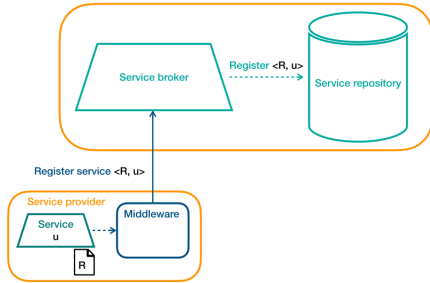
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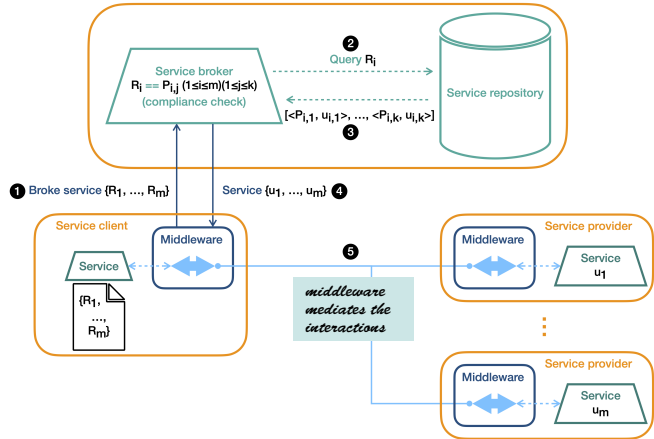
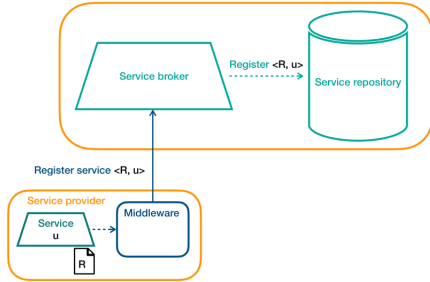
SEARCH basics



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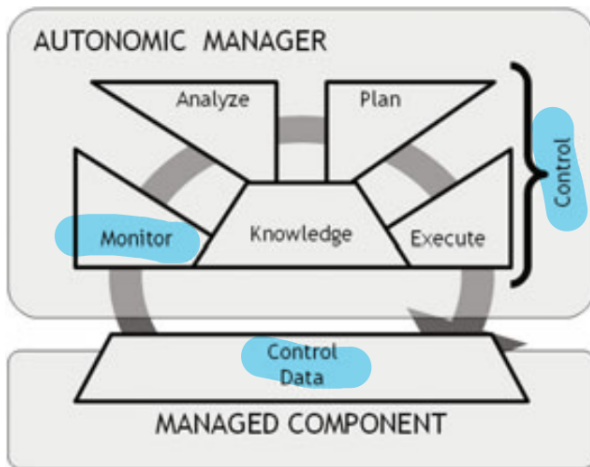


SEARCH basics



– On adaptation –

Adaptation basics



Components' behaviour = control + data

- ▶ *adaptation = process/modify specific computational/physical resources*
- ▶ *control data [2]^a*
 - ▶ *memory usage*
 - ▶ *network traffic*
 - ▶ *room temperature*
 - ▶ *...*

*Control data are an
"adaptation"-dependent design choice*

^aPicture borrowed from [2]

A benchmark for adaptation

Bliudze et al.

5

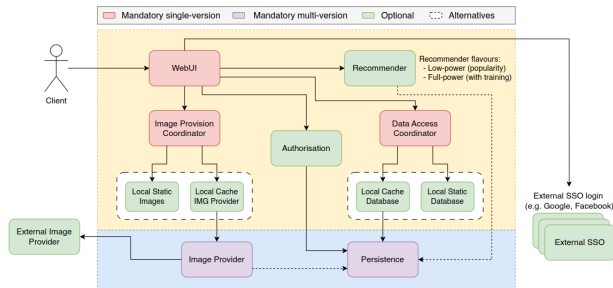


Figure 2: Adaptable TeaStore Architecture (functionalities shown within the coloured areas are internal, those in the blue area are outsourceable, in the yellow area—non-outsourceable (cf. Table I))

“a broad catalogue of reference adaptation scenarios centred around Adaptable TeaStore, useful to evaluate the ability of a given adaptation technology to address conditions such as component unavailability, cyberattacks, provider outages, benign/malicious traffic increases, and user-triggered reconfigurations” [1]

A running example

¹Adapted from [1, 7]

A running example

The adaptable TeaStore benchmark¹

- ▶ a webUI (app users' entry point)
- ▶ a remote image provider
- ▶ a locally hosted DB of images

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A running example

The adaptable TeaStore benchmark¹

- ▶ a webUI (app users' entry point)
- ▶ a remote image provider
- ▶ a locally hosted DB of images

According to the specification

- ▶ providers process images (e.g., applying filters, resizing, etc.);
- ▶ webUI automatically generates webpages with images
 - ▶ from image providers
 - ▶ or from DB for runtime adaptation
 - ▶ or else directly from the hard drive

¹Adapted from [1, 7]

Our adaptable TeaStore

Our adaptable TeaStore

Adaptation triggered by service degradation depending on

- ▶ network L atency
- ▶ R esponsiveness of image provider
- ▶ C PU load of webUI

$$\mathcal{E}_1 = \forall x : L < x \implies R < x^2$$

$$\mathcal{E}_2 = \neg \mathcal{E}_1 \quad \wedge \quad 0 \leq C \leq .9$$

$$\mathcal{E}_3 = \neg \mathcal{E}_1 \quad \wedge \quad C > .9$$

Our adaptable TeaStore

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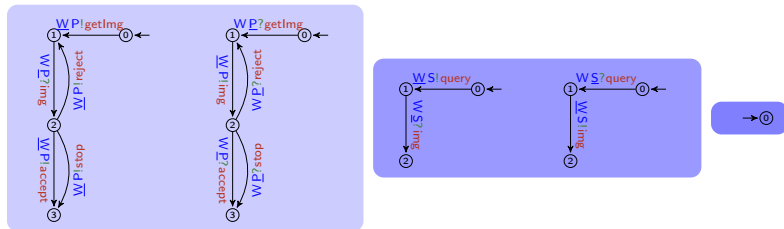
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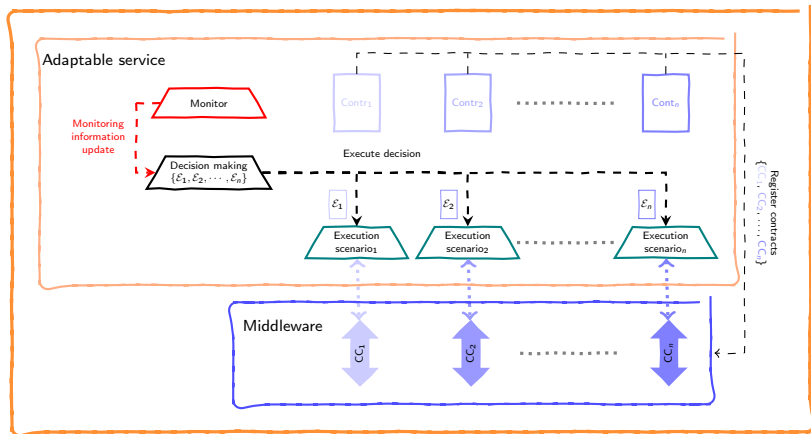
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Contracts for the 3 scenarios:

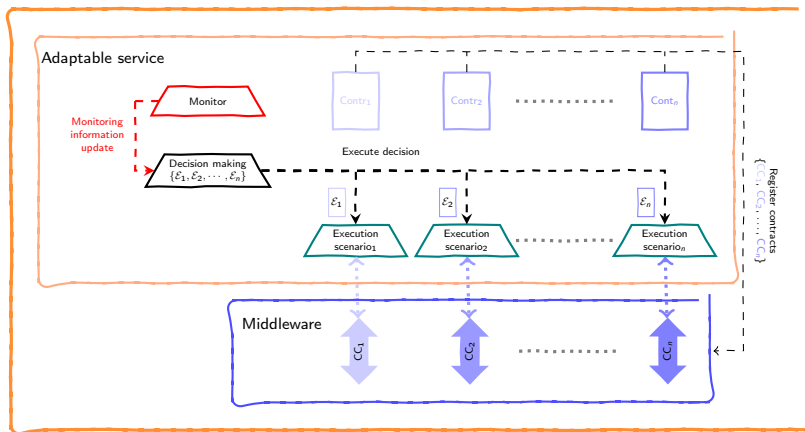


– Adaptation as service composition –

Architecture of our adaptation pattern



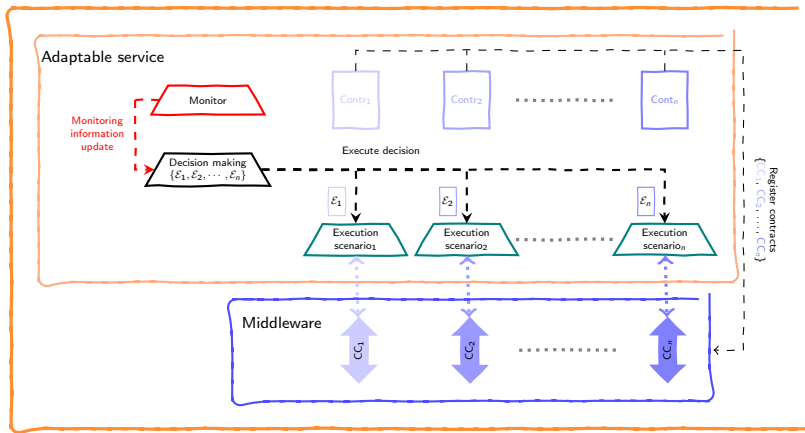
Architecture of our adaptation pattern



Middleware

- ▶ mediates interactions
- ▶ through ports' connections

Architecture of our adaptation pattern



Adaptable service

- ▶ a contract for each channel
- ▶ execution scenarios imp.
- ▶ adaptation policy

Middleware

- ▶ mediates interactions
- ▶ through ports' connections

Control loop

while ...:

m = monitor's last update of control data

next_scenario = choose(m)

if next_scenario $\neq$ running_scenario:

stop running_scenario

trigger next_scenario

$$m : \begin{cases} L \mapsto 2 \\ R \mapsto 3 \\ C \mapsto .5 \end{cases}$$

and e.g., rely on *RC7* [6, 4, 3]
which uses a *SM7* solver

A control loop as “service combinator”

```
1 from threading import Thread
2 from ExcScenarios import ES, done
3
4 done = False
5
6 while <not done>:
7     stop, scenario = False, None
8     i = 0
9     while scenario == None:
10         m = monitor_update()
11         if not SMT(not  $\mathcal{E}_i[a \mapsto m(a)]_{a \in \mathcal{A}}$ ) == SAT:
12             scenario = Thread(target = ES[i])
13             scenario.start()
14         else:
15             i = (i + 1) mod len(ES)
16
17 while not (done or stop):
18     old_m = m
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ES *list of functions each implementing an execution scenario*

done *shared for coordinated termination*

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*spawn the chosen scenario ES[i],
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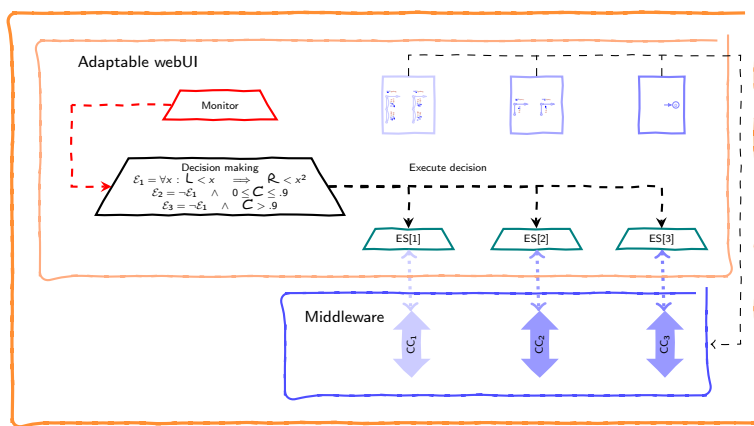
get control data

*spawn the chosen scenario $ES[i]$,
if enabled*

*keep going until the scenario is done or
adaptation is needed*

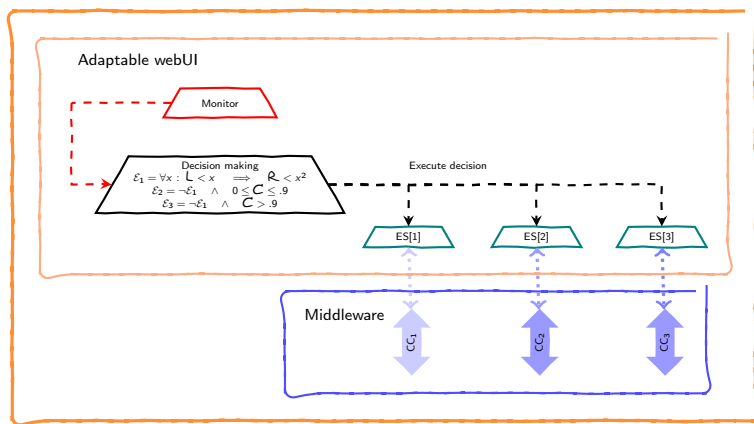
– Our pattern at work –

The pattern in action

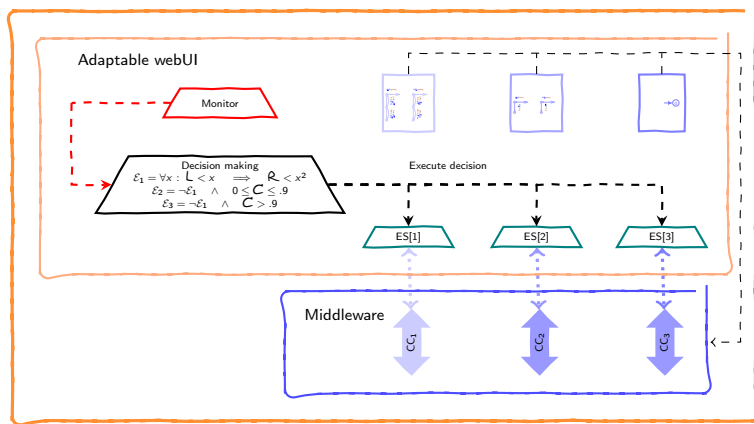


*Comm. contracts registered
on local middleware*

The pattern in action

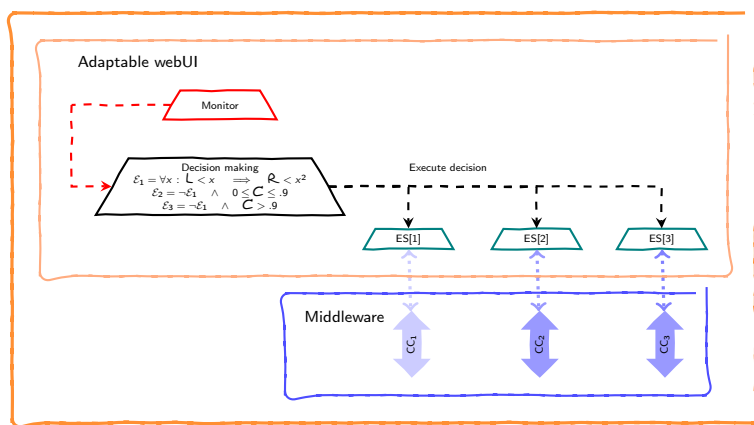


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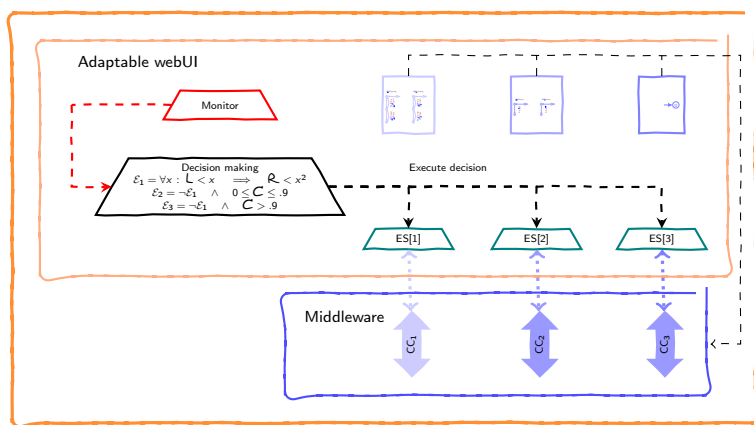


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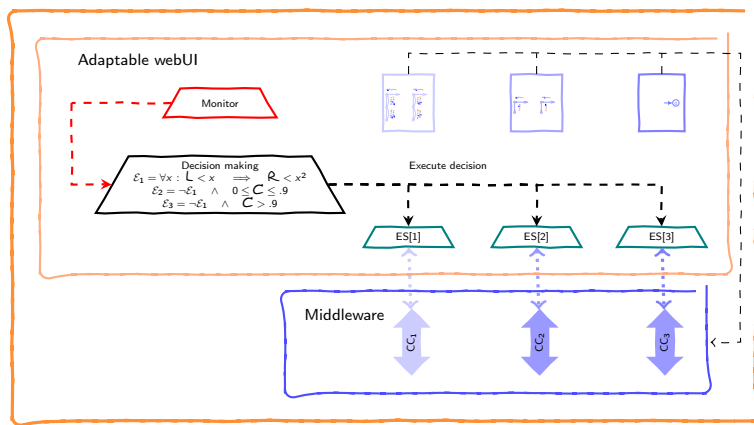


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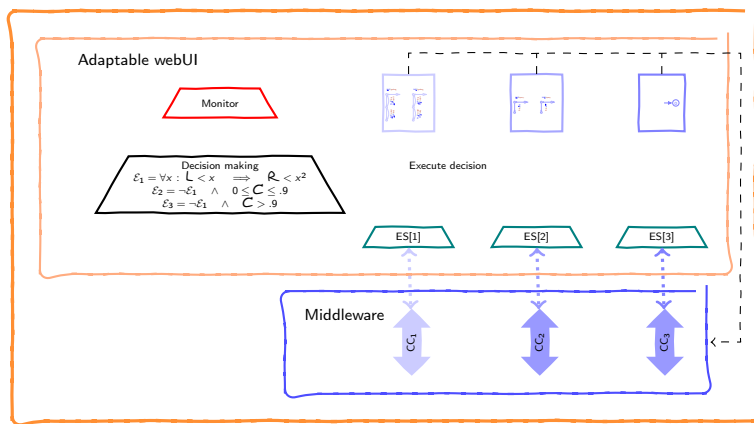
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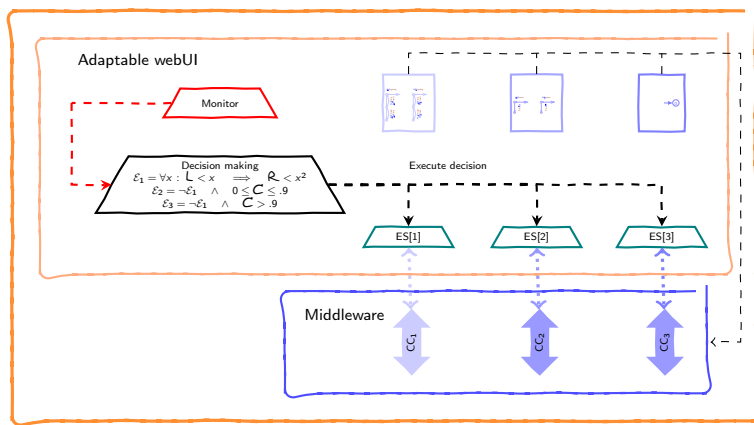
*Monitor and decision
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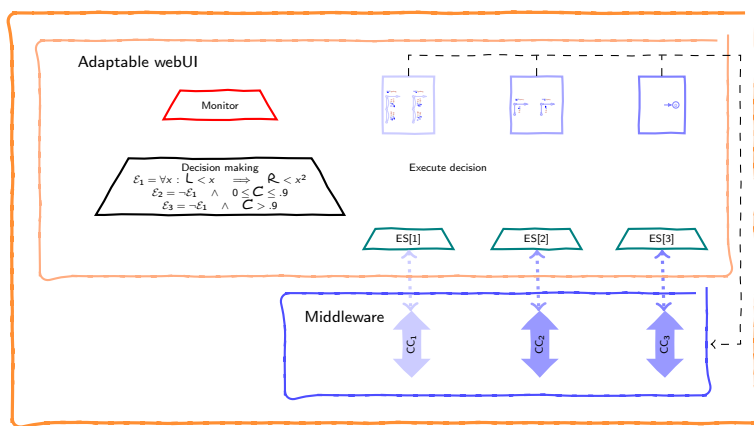
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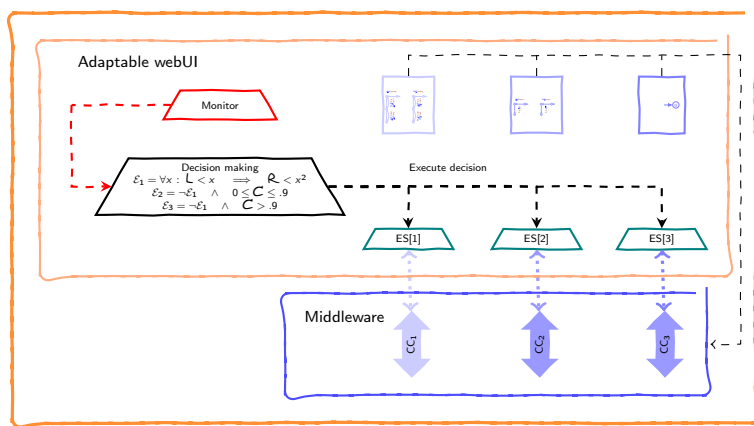
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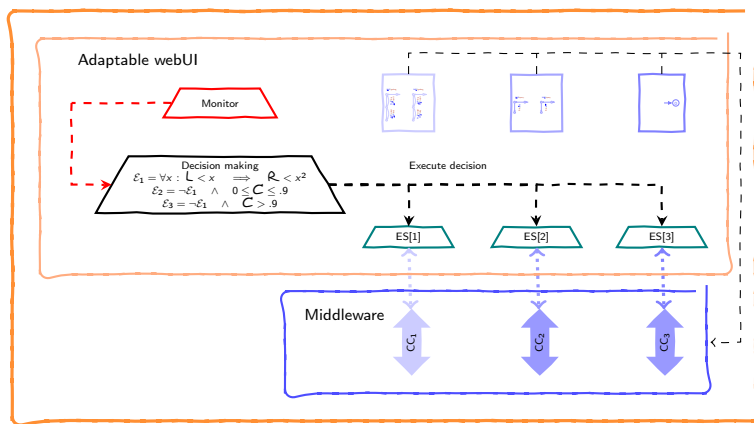
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Control data acquired:
 $L \mapsto 2, R \mapsto 3, C \mapsto 0.54$

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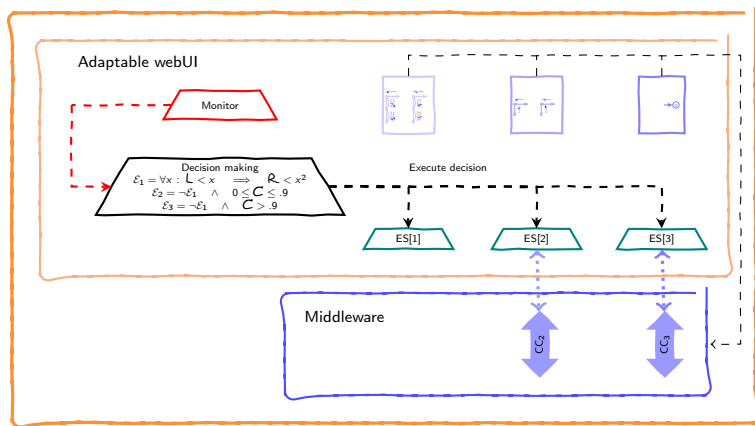
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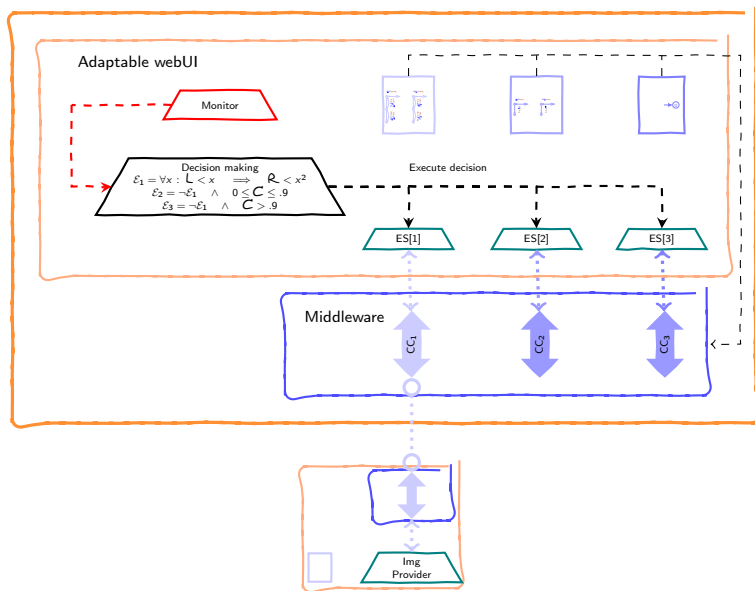


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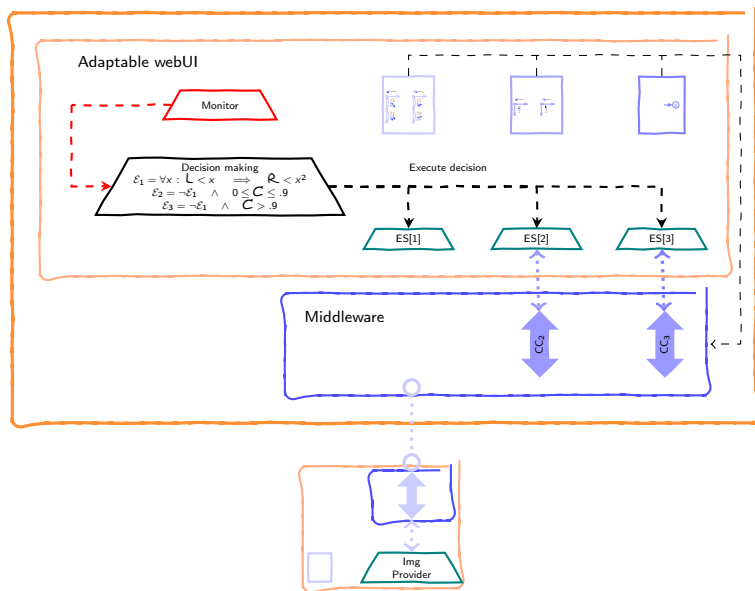
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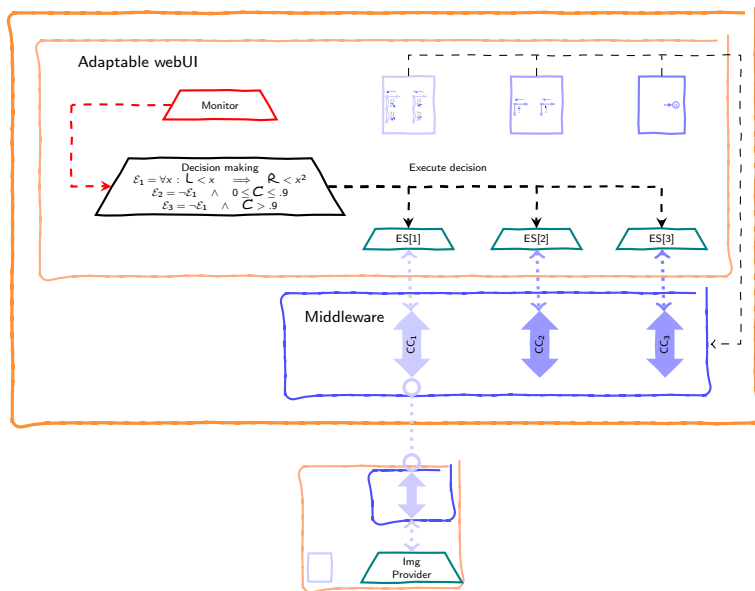
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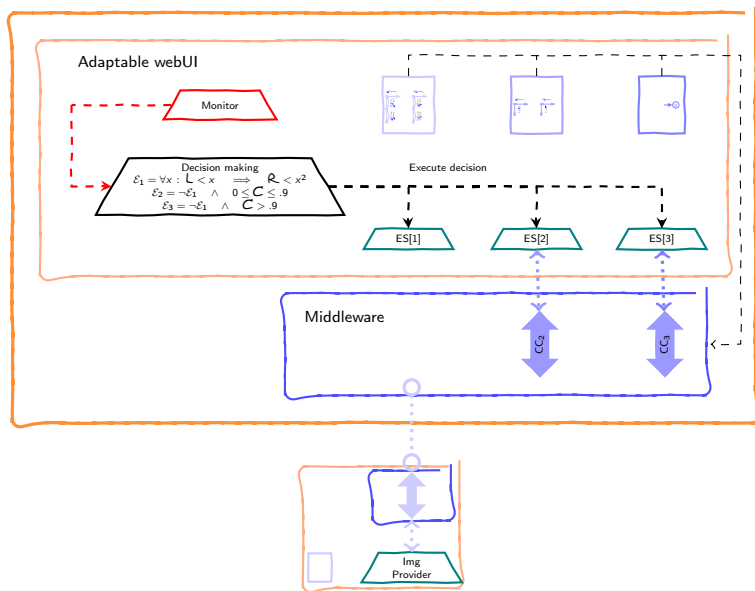
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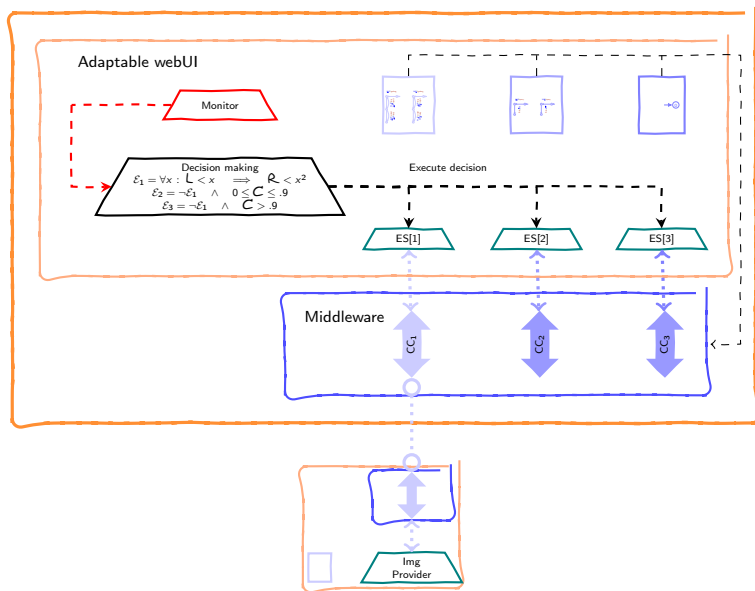
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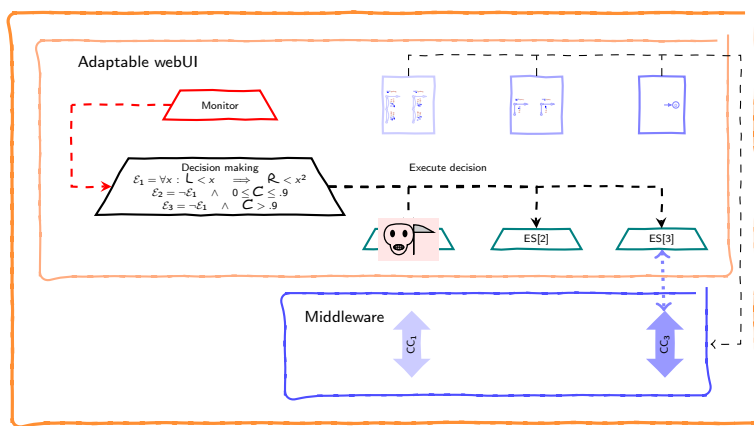


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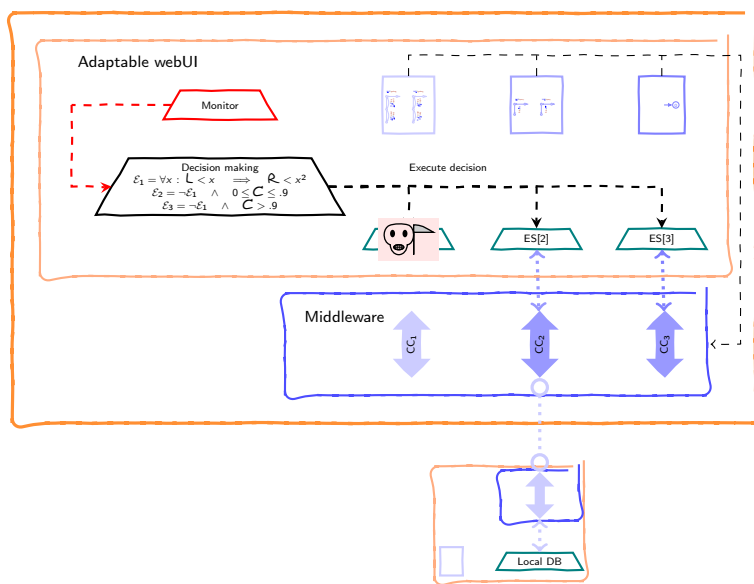
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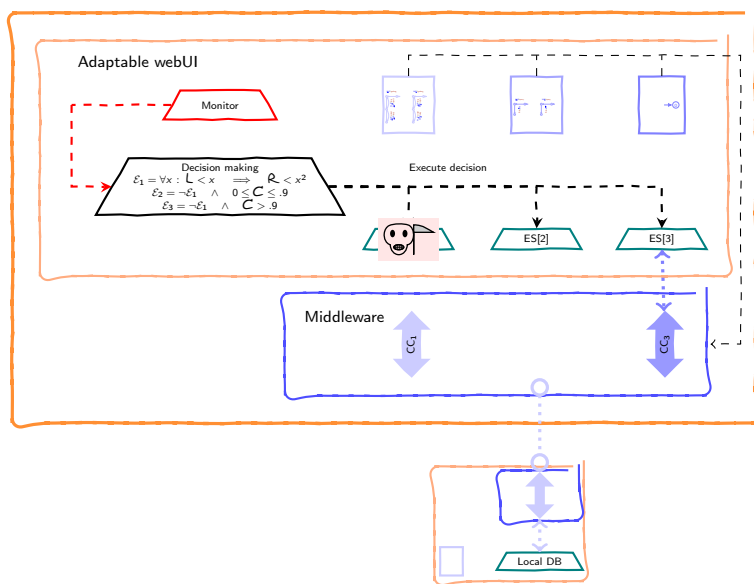
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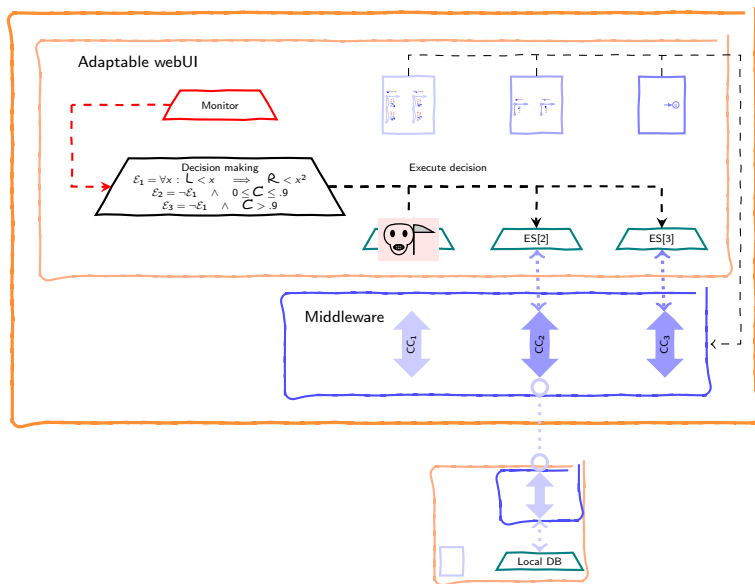


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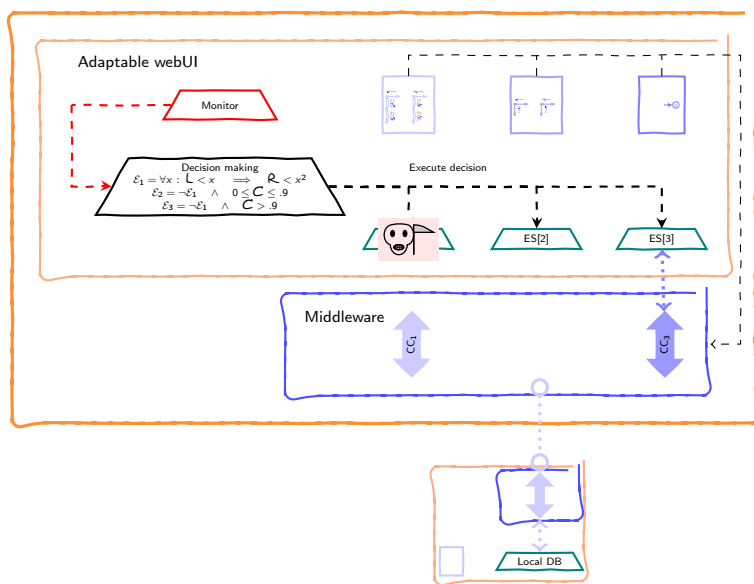


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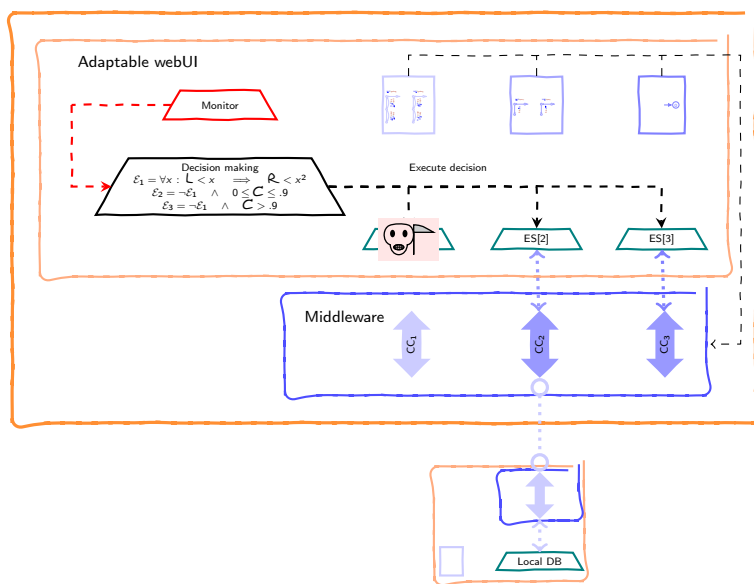
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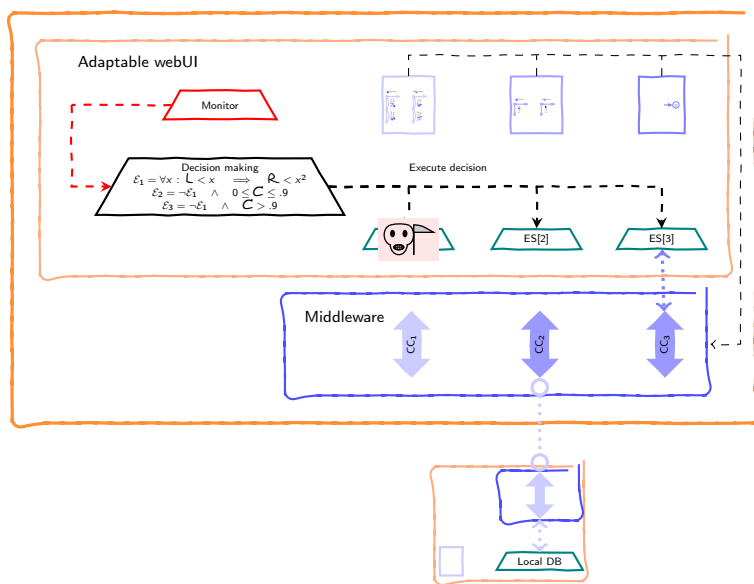
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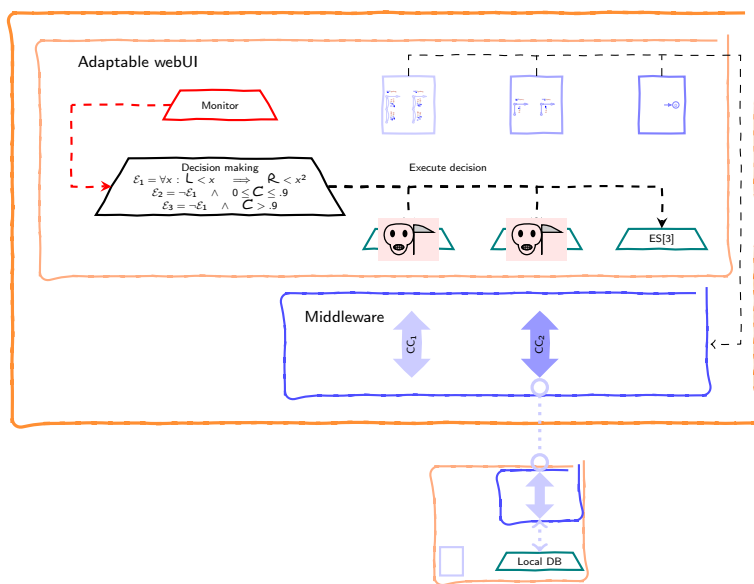
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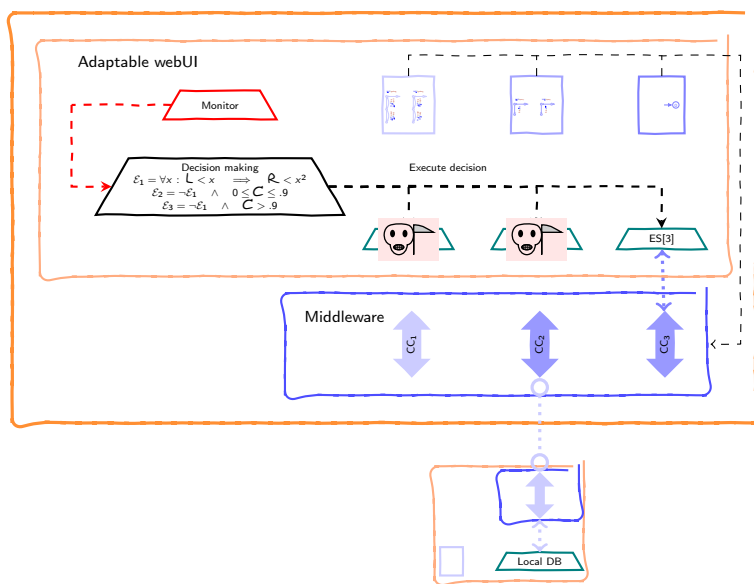
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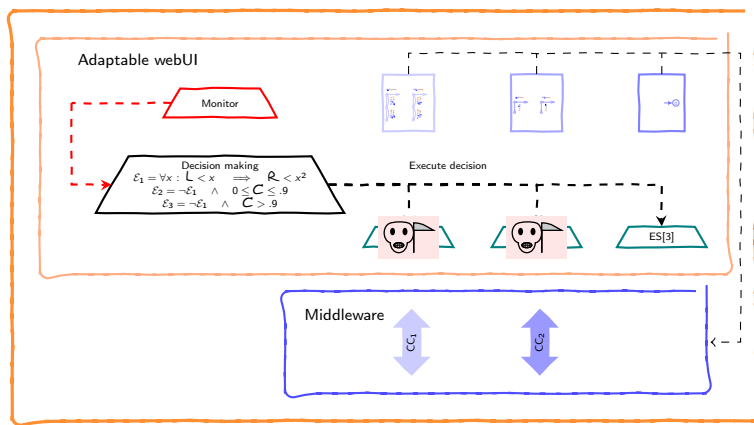
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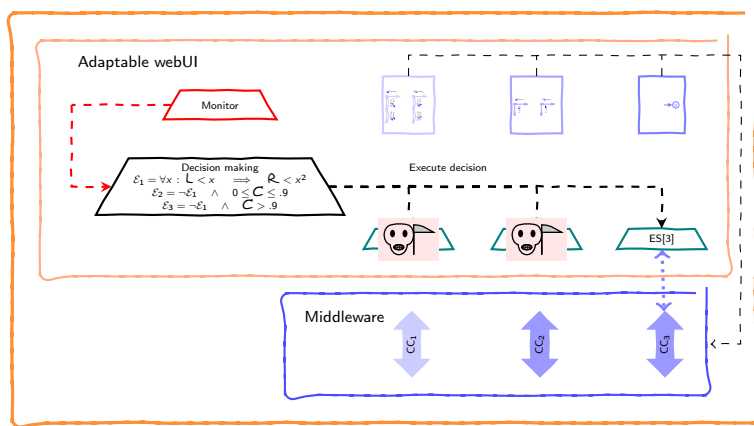
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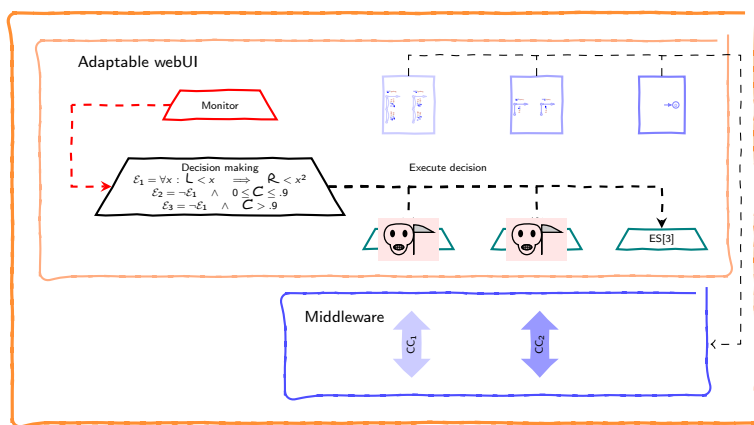
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The pattern in action



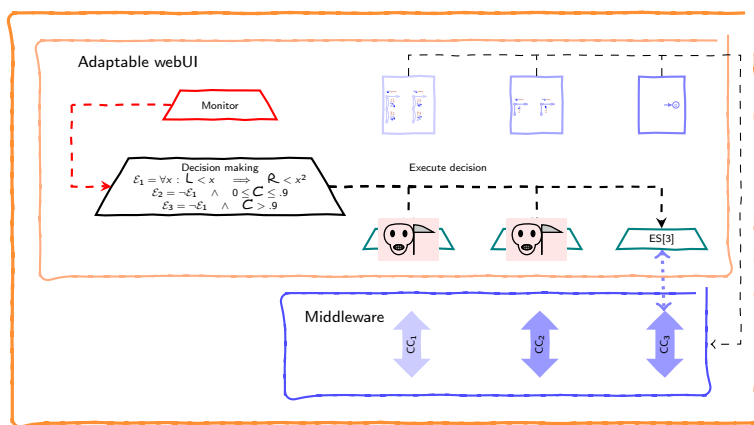
Control data acquired:
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– Epilogue –

Wrapping up

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- ▶ Take a look at the paper for
 - ▶ considerations about the control loop
 - ▶ details about adaptation of the running example

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- ▶ more general combinators (e.g., monitor passing data to execution scenarios)
- ▶ abrupt termination
 - ▶ move to graceful adaptation (e.g., ensuring transactional behaviour)

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R3: Presentation

- Apologies (it was a rush)...thank for the great help in improving the presentation

Thank you!

References I

- [1] Simon Bliudze, Giuseppe de Palma, Saverio Giallorenzo, Ivan Lanese, Gianluigi Zavattaro, and Brice Arleon Zemtsop Ndadji. Adaptable TeaStore.
On-line, 2024. Available at <https://arxiv.org/abs/2412.16060>.
- [2] Roberto Bruni, Andrea Corradini, Fabio Gadducci, Alberto Lluch Lafuente, and Andrea Vandin. A conceptual framework for adaptation. In Juan de Lara and Andrea Zisman, editors, *Fundamental Approaches to Software Engineering*, pages 240–254, Berlin, Heidelberg, 2012. Springer Berlin Heidelberg.
- [3] Carlos G. Lopez Pombo, Agustín Eloy Martinez Suñé, and Emilio Tuosto. A dynamic temporal logic for quality of service in choreographic models.
In Erika Ábrahám, Clemens Dubslaff, and Silvia Lizeth Tapia Tarifa, editors, *Proceedings of 20th International Colloquium on Theoretical Aspects of Computing - ICTAC 2023*, volume 14446 of *LNCS*, pages 119–138, Lima, Perú, December 2023. Springer.
- [4] Carlos G. Lopez Pombo, Agustín Eloy Martinez Suñé, and Emilio Tuosto. Automated static analysis of quality of service properties of communicating systems.
In André Platzer, Kristin Yvonne Rozier, Matteo Pradella, and Matteo Rossi, editors, *Proceedings of 26th. International Symposium on Formal Methods (FM 2024) - Part II*, volume 14934 of *LNCS*, pages 84–103. Springer, September 2025.
- [5] Carlos Gustavo López Pombo, Pablo Montepagano, and Emilio Tuosto. Search: An execution infrastructure for service-based software systems.
In Ilaria Castellani and Francesco Tiezzi, editors, *Coordination Models and Languages - 26th IFIP WG 6.1 International Conference, COORDINATION 2024, Held as Part of the 19th International Federated Conference on Distributed Computing Techniques, DisCoTec 2024, Groningen, The Netherlands, June 17-21, 2024, Proceedings*, volume 14676 of *Lecture Notes in Computer Science*, pages 314–330. Springer, 2024.
- [6] Carlos López Pombo, Agustín E. Martinez Suñé, and Emilio Tuosto. A dynamic temporal logic for quality of service in choreographic models. *TCS*, 1043:115247, 2025.

- [7] Jóakim von Kistowski, Simon Eismann, Norbert Schmitt, André Bauer, Johannes Grohmann, and Samuel Kounev. Teastore: A micro-service reference application for benchmarking, modeling and resource management research.
In *26th IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS 2018)*, pages 223–236. IEEE Computer Society, September 2018.